Example 4.2

When a number is decreased by 25%, it becomes 120. Find the number.

Solution:

Let the number be x.

Given, $x - \frac{25}{100}x = 120$

$$\frac{100x - 25x}{100} = 120$$

$$\frac{75x}{100} = 120$$

$$\Rightarrow x = \frac{120 \times 100}{75}$$

$$\Rightarrow x = 160$$



If we start with a quantity A and then decrease that quantity by x%, we will get the decreased quantity as,

$$D = \left(1 - \frac{x}{100}\right) A$$

Aliter: Using the above formula,

$$D = \left(1 - \frac{25}{100}\right)A \Rightarrow 120 = \frac{75}{100} \times A$$

$$A = 120 \times \frac{100}{75} = 160$$

Example 4.3

Akila scored 80% of marks in an examination. If her score was 576 marks, then find the maximum marks of the examination.

Solution:

Let the maximum marks be x.

Now, 80% of $x = 576$

$$\frac{80}{100} \times x = 576$$

$$\Rightarrow x = 576 \times \frac{100}{80} = 720 \text{ marks}$$

Therefore, the maximum marks of the examination = 720.

Example 4.4

If the price of Orid dhall after 20% increase is ₹96 per kg, then find the original price of Orid dhall per kg.

Solution:

Let the original price of Orid dhall be ₹x.

New price after 20% increase

$$= x + \frac{20}{100}x = \frac{120x}{100}$$

Given that, $\frac{120x}{100} = 96$

$$\therefore x = \frac{96 \times 100}{120}$$

$\therefore$ Original price of Orid dhall per kg, x = ₹80



If we start with a quantity A and then increase that quantity by x%, we will get the increased quantity as,

$$I = \left(1 + \frac{x}{100}\right) A$$

Aliter: Using the above formula,

$$I = \left(1 + \frac{20}{100}\right)A$$

$$\Rightarrow 96 = \frac{120}{100} \times A$$

$$A = 96 \times \frac{100}{120}$$

A = ₹80

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